3x26 = 78

0000000000000011
00000000

Coal Assignment # 2

Date _____

Q1. 26 = 16 + 8 + 2
= $2^4 + 2^3 + 2^1$

```
.code
mov ax, 3
mov bx, ax
mov cx, ax

Shl ax, 4
Shl bx, 3
Shl cx, 1

add, ax, bx
add ax, cx.
```

Q2.
```
.code
mov edx, 02345ABCDh

rol edx, 8
rol edx, 8
rol edx, 8
rol edx, 8
```

Q2.
```
mov eax, edx
call WriteHex
call CRLF
ror edx, 8
ror edx, 8
ror edx, 8
ror edx, 8
mov eax, edx
call WriteHex
call CRLF.
```

Q3.
```
.code
mov edx, 02345ABCDh
rol edx, 16
rol edx, 16
mov eax, edx
ror edx, 16
ror edx, 16
mov eax, edx.
```

Q4.

a.
```
.data
BSeconds BYTE ?
BMinutes BYTE ?

.code
mov dx, 0101 1001 1101 1010
mov ax, dx
and ax, 0000 0000 0001 1111
mov Bseconds, al
```

b.
```
mov ax, dx
shr ax, 5
and ax, 0000 0011 1111
mov BMinutes, al.
```

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9A3F
1
9A40
1
9A41

~~7F80~~
+90EA
106A

16 | 22
-16
6

Date _____

Q5.

106A
8120
918A

9A41
FFFF
9A40

Step 1 → ax = 106Ah, CF = 1, dx = 9A3Fh

Step 2 → dx = 9A41h, ax = 106Ah, CF = 0

Step 3 → ax = 918Ah, dx = 9A41, CF = 0

Step 4 → ax = 918Ah, dx = 9A40, CF = 1

Q6. a. test ax, ax

b. test bx, 1

c. test dx, 8000h

d. test dx, 8000h

e. test bl, 0FFh

Q7. mov dx, 5 ⇝ dx = 5

stc ⇝ (CF = 1)

mov ax, 10h ⇝ (AX = 16)

adc dx, ax ⇝ dx = 16 + 5 + 1 = 22

dx = (0016h)

Q8.

ERRORS		CORRECTION
mov cx, 8	→	mov ecx, 8
mov BYTE PTR [esi], al	→	mov BYTE PTR [ebx], al
		(*) where mov ebx, OFFSET result
dec esi	→	inc esi
dec edi	→	inc edi

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Q9.

```
.data
marks BYTE ?
msg BYTE "Input marks", 0
.code
mov edx, OFFSET msg
call WriteString
call ReadInt
mov marks, eax
cmp eax, 80
jge GradeA
cmp eax, 60
jge GradeB
cmp eax, 40
jge GradeC
mov edx, OFFSET m5
jmp display

GradeA:
mov edx, OFFSET m1
jmp display
GradeB:
mov edx, OFFSET m2
jmp display
GradeC:
mov edx, OFFSET m3
jmp display
display:
call writeString
call crlf
exit
main ENDP
END main
```

```
m1 BYTE "Grade A", 0
m2 BYTE "Grade B", 0
m3 BYTE "Grade C", 0
m4 BYTE "Grade D", 0
m5 BYTE "Fail", 0
```

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Date _____

Q10.

	EAX	EBX	Stack
mov eax, 3	3		-
mov ebx, 7	3	7	-
push eax	3	7	3 
push ebx	3	7	3 then 7 
pop eax	7	7	3 
push 10	7	7	3 then 10 
pop ebx	7	10	3
pop eax	3	10	-

Q11.

```
.data
num1 SDWORD ?
num2 SDWORD ?
num3 SDWORD ?
num4 SDWORD ?
result SDWORD ?
.code
MAIN PROC
mov eax, num1
~~mov ebx, num2~~
cmp eax, num2
jle elseif
    mov ebx, num3
    cmp ~~num3~~ ebx, num4
    je elseit
    ~~mov eax, ebx~~
    add eax, ~~ebx~~ ebx
    mov result, eax
jmp finish

elseif:
    mov eax, num2
    cmp num2, num4
    jne else
    ~~sub~~ sub eax, num1
    mov result, eax
    jmp finish

else:
    mov result, 0

finish:
    ; call writedec // any remaining code
    exit
main ENDP
END main
```

op1 op2

Date _____

Q12.

```
.data
op1 SDWORD ?
op2 SDWORD ?
x   SDWORD ?
y   SDWORD ?
z   SDWORD ?

.code
MAIN PROC
While :
    mov eax, op1
    cmp eax, op2
    jg EndWhile
    cmp eax, x
    jl else
    cmp eax, y
    jg else
    add z, 10
    jmp decrement
Else :
    sub z, 10
    jmp decrement
Decrement :
    dec op--
    jmp while
EndWhile
    // remaing code.
    exit
main ENDP
END MAIN
```

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